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Key Points:
- New paleomagnetic data from Oligocene volcanoes in Australia improve its apparent polar wander path
- Our updated Cenozoic apparent polar wander path for Australia indicates acceleration after 34 Ma
- Elongation of site mean magnetic directions along the Cosgrove track empirically validates TK03.GAD

Supporting Information:
- Supporting Information S1
- Data Set S1

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Paleomagnetism of Oligocene Hot Spot Volcanics in Central Queensland, Australia

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Abstract We report three high-quality paleomagnetic poles for Oligocene Australia from volcanic provinces along the Cosgrove hot spot track in central Queensland, NE Australia. We present paleomagnetic poles for Springsure (27.3–28.9 Ma: $\phi = 300.6°$E, $\lambda = 70.5°$N, $K = 13.7$, $A_{95} = 9.7°$, $N = 18/22$, $Q = 6$), Peak Range (28.2–31.2 Ma: $\phi = 291.8°$E, $\lambda = 64.6°$N, $K = 10.1$, $A_{95} = 8.8°$, $N = 29/31$, $Q = 6$), and Hillsborough volcanic provinces (33.1–34.1 Ma: $\phi = 301.4°$E, $\lambda = 67.1°$N, $K = 32.2$, $A_{95} = 7.1$, $N = 14/18$, $Q = 7$), as well as data from Buckland and Nebo volcanic provinces. Using our new results, we resolve ambiguities present in published paleomagnetic data for Australia during the Oligocene epoch (23–34 Ma) and calculate a combined Oligocene paleomagnetic pole of $\phi = 289.2°$E, $\lambda = 71.0°$N, ($K = 24.6$, $A_{95} = 1.9°$, $N = 227/274$, $Q = 6$). We update the continental apparent polar wander path for Australia and find evidence for accelerated seafloor spreading in the Southern Ocean after ~ 34 Ma (anomaly C13). Lastly, we take advantage of the large number of paleomagnetic sites ($N = 97$) along the Cosgrove hot spot track to empirically evaluate the TK03.GAD geomagnetic field model. Paleomagnetic directions from rocks along the Cosgrove track have an elongation (E) of 1.6, remarkably consistent with TK03.GAD expected values for hot spot paleolatitudes ($E = 1.4$–1.6, $\lambda = 36.6$–$43.3°$S).

1. Introduction

The purpose of this study is to present modern paleomagnetic data from five Oligocene-aged volcanic provinces in central Queensland, in northeastern Australia, and to provide a synthesis with existing data for Australia from this epoch. Although the earliest paleomagnetic results from Australia no longer meet modern criteria for paleomagnetic data (samples were not demagnetized; Green & Irving, 1958; Irving & Green, 1958; Irving et al., 1961), many studies from the Cenozoic era that were conducted since have fallen out of use due to outdated blanket demagnetization techniques, leaving gaps in the continental paleomagnetic record. While some validation of these results has been accomplished for parts of the Cenozoic era (specifically a revision of Newer Volcanic Province results, Aziz-Ur-Rahman, 1971; Opdyke & Musgrave, 2004; and validation of earlier Barrington Volcano, Hill et al., 2002; Wellman et al., 1969; Nandewar Volcano and the Port Campbell and Glenample Formations results, Hansma & Tohver, 2018; Idnurm, 1985; Wellman et al., 1969), Oligocene results remain to be examined. A dramatic increase in Australia's northward drift at the beginning of the Oligocene (~ 34 Ma) is indicated by seafloor magnetic anomaly patterns in the Southern Ocean, with acceleration occurring after anomaly C13 (Tikku & Cande, 1999). Global plate models predict a doubling of Australia's apparent polar wander (APW) rate after C13, from ~ 0.3°/Ma between 60 and 34 Ma to ~ 0.6–0.7°/Ma between 34 and 0 Ma (Matthews et al., 2016; Seton et al., 2012). However, paleomagnetic documentation of this transition to rapid drift rates is not clear from existing apparent polar wander paths (APWPs; Idnurm, 1985, 1994; McElhinny et al., 1974; Musgrave, 1989).

While published Oligocene paleomagnetic results for Australia exist, the seven igneous results are blanket demagnetized, whereas stepwise demagnetization is limited to a pair of results from sedimentary rocks, preventing evaluation of seafloor spreading model predictions (Idnurm, 1985, 1994; McElhinny et al., 1974; Mumme, 1963; Robertson, 1966; Wellman, 1975; Wellman et al., 1969). Excessive filtering of blanket-demagnetized samples was identified by Musgrave (1989) as the cause of unrealistic zigzags in the APWP of McElhinny et al. (1974). Prior to this realization, however, these deviations falsely implied episodic volcanism and, by extension, inadequate sampling of paleosecular variation (PSV). As such, focus shifted to generating sedimentary paleomagnetic results, culminating in an APWP that rejected almost all paleomagnetic data from igneous rocks (Idnurm, 1985, 1994), with the exception of the Newer Volcanics data

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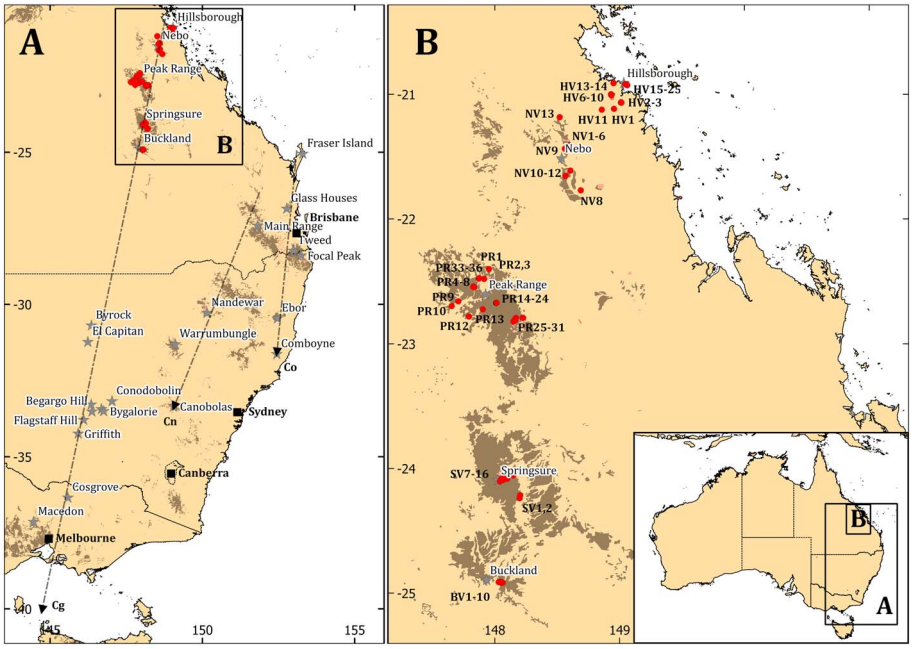


Figure 1. Onshore Cenozoic magmatic provinces, in (a) eastern Australia and (b) central Queensland. Approximate paths of the Cosgrove (Cg), Canobolas (Cn), and Comboyne (Co) hot spot tracks are indicated with dashed lines; central volcanic provinces are indicated by gray stars; and sampling sites are indicated by red points. BV = Buckland Volcano; SV = Springsure Volcano; PR = Peak Range; NV = Nebo Volcano; HV = Hillsborough Volcano.

set (Aziz-Ur-Rahman, 1971). It is somewhat surprising that earlier igneous data were so readily rejected, especially given the consistency of virtual geomagnetic pole (VGP) samples across each of these studies. Musgrave (1989) introduced a third APWP contender by combining igneous and weathering data together. Aside from the circularity of using undated paleomagnetic poles from laterites to define the APWP, the greatest shortcoming of this path is its constant APW rate. Perhaps more significant than development of paleomagnetic methods has been improvements for the age of Cenozoic volcanics in eastern Australia associated with the adoption of mineral separate $^{40}Ar/^{39}Ar$ geochronology over whole rock K-Ar methods (Ashley et al., 1995; Cohen et al., 2007; Cohen, Knesel, et al., 2013; Knesel et al., 2008; supporting information Table S1). As such many of the magnetization ages related to paleomagnetic results need to be updated before reconciliation of sedimentary and igneous paleomagnetic results can be realized.

2. Geological Background and Sampling

Remnants of Oligocene hot spot volcanism are exposed along a ~ 450-km-long north-south linear chain in central Queensland from Cape Hillsborough to Buckland Volcano (Figure 1). The rocks were generated by the Cosgrove hot spot, that, along with leucitite lavas preserved in New South Wales and Victoria, forms the longest continental hot spot track on Earth (Davies et al., 2015; Sutherland, 1981). The Cosgrove track is the westernmost of five southward younging Cenozoic hot spot tracks on the eastern Australian Plate, with Comboyne and Canobolas hot spots having produced two other onshore hot spot tracks and the Tasmantid and Lord Howe hot spots having produced two offshore tracks. Finite-frequency tomography indicates that the Cosgrove hot spot itself lies ~ 300 km beneath northeastern Tasmania and the Bass Strait (Montelli et al., 2006).

The central Queensland segment of the Cosgrove track is further divided into five progressively younger volcanic provinces that represent volcanic activity from 34 Ma at Hillsborough Volcano, followed by Nebo (33–32 Ma), Peak Range (31–28 Ma), Springsure (28.5–27.5 Ma), and Buckland Volcanoes (28–27 Ma; Cohen, Knesel, et al., 2013). The volcanos in each province are shields of chemically bimodal composition, consisting of variably eroded basaltic lavas, accompanied by late stage silicic volcanism.

Basalts have tholeiitic composition and commonly contain olivine and plagioclase phenocrysts, set in a groundmass also containing pyroxene, and accessory titanomagnetite, ilmenite, and apatite. Silicic rock

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compositions range from trachytic to rhyolititc, common phenocrysts including anorthoclase-sanidine, sodic plagioclase, and quartz. Biotite, hornblende, titanomagnetite, ilmenite, and zircon are common accessory phases. Chemical weathering since eruption has in some localities produced Liesegang bands of Fe oxides (hematite) in silicic magmas. Trace element and isotopic compositions indicate an ocean island basalt-type, metasomatized mantle source (EM-1 type) with little crustal assimilation (Davies et al., 2015; Ewart, 1982).

The extrusive volume of each central Queensland volcanic province has been estimated at ~ 500 km^3, with perhaps twice as much volume of intrusive material (Wellman & McDougall, 1974). Thinner lithosphere and melting at shallower crustal depths (< 110 km) led to significantly greater volumes of melt production for the Cosgrove hot spot in Queensland than in New South Wales, where greater lithospheric thickness restricted melting, producing leucitites lavas (Davies et al., 2015).

2.1. Sampling

Multiple oriented samples (7–10 cores) were collected from outcrop using a gasoline-powered Pomeroy drill at 94 individual sites, that is, cooling units (81 of which produced consistent within-site magnetizations). Basaltic sites were collected from roadcut and streambed exposures, as well as incised drainage beneath silicic capped mesas. Silicic rocks were also taken from roadcuts, as well as at the base of large, natural exposures. Site location was recorded with handheld GPS, and sample orientation was taken by magnetic and solar compass. Magnetic orientations were corrected using WMM2015-2025 (Chulliat et al., 2015) and, in most cases, corroborated solar azimuths, differing by < 5°. In the few cases where local magnetic azimuths deviated by greater amounts, solar azimuths were given preference. Three pairs of sites, PR18 and PR19, PR20 and PR21, and HV16 and HV17, were taken from the same flow/felsic intrusion, and samples were combined. Of the five volcanic provinces sampled, Springsure is the only one where previous paleomagnetic work has taken place, comprising seven sites from Mount Boorambool (McElhinny et al., 1974).

2.2. Buckland

Buckland Volcano is an ~ 80-km-wide shield volcano erupted onto Jurassic sandstones. Clear radial drainage is preserved (Young & Wray, 2000), and ~ 300 m of horizontal basaltic lava thickness remains, forming tableland topography (Skae, 1998). Flows are composed of basalts, trachybasalts, and highly alkaline basanitic lavas, with three minor felsic intrusions found near the center of the volcano (Skae, 1998). Another group of alkali mafic plugs and flows lie ~ 60 km SW of the main volcanic center (Mollan et al., 1969; Skae, 1998). An $^{40}Ar/^{39}Ar$ total fusion age of 27.3 ± 0.4 Ma from sanidine of peralkaline trachyte plug (Cohen, Knesel, et al., 2013) is compatible with fission track age of 27.6 ± 1.1 Ma (Hollis & Sutherland, 1985). Ten paleomagnetic sites of basaltic flows (BV1–BV10) were sampled (Figure 1).

2.3. Springsure

Springsure volcano sits atop Lower Permian sandstones and has a central zone of silicic plugs and flows, the Minerva Hills Volcanics, surrounded by mesas and plains of mafic lava (Cohen, Withnall, et al., 2013). Lava flows have a maximum thickness of ~ 330 m near the volcano center, with lavas being broadly divided into lower mafic flows, interbedded mafic and silicic flows that include the Minerva Hills plugs, and upper mafic flows (Mollan et al., 1969; Sutherland et al., 1989). The flows include olivine basalt and less basic, intermediate, lavas. The pyroclastics near Springsure consist of massive reddish purple scoriaceous tuffs of trachytic composition and well-bedded off-white fine rhyolitic agglomerate and tuff (Mollan et al., 1969; Sutherland et al., 1989). The Minerva Hills Volcanics consist of a suite of alkaline rocks ranging from trachyte to alkaline rhyolite. They are preserved mainly as dykes, plugs, and domes, but a dissected series of flows are also present. An $^{40}Ar/^{39}Ar$ total fusion analysis of sanidine from a silicic intrusion gave an age 28.1 ± 0.3 Ma (Cohen, Knesel, et al., 2013). Fifteen paleomagnetic sites of mafic and silicic units were sampled from this province that, together with seven previously published sites from Mount Boorambool (McElhinny et al., 1974), comprise the 22 sites sampled.

2.4. Peak Range

The Peak Range volcanic field stretches over 100 km in a N-S direction with older, silicic plugs in the north (30.7 ± 0.4 and 30.3 ± 0.4 Ma), mesas of mafic lavas in the center (30.4 ± 0.5 Ma), and younger silicic plugs in the south (29.0 ± 0.4 and 28.6 ± 0.4 Ma; Cohen, Knesel, et al., 2013; Cohen, Withnall, et al., 2013). The northern plugs are peraluminous rhyolites, while the southern plugs are peralkaline rhyolites and trachytes (Cohen, Withnall, et al., 2013). Surrounding the plugs are mafic lavas, commonly covered by soil but exposed in riverbeds and roadcuttings. In central Peak Range, the residual mesas of Lord's Table Mountain attest to

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the volume of eroded mafic material, preserving up to 50 flows with a thickness of 300 m (Cohen, Withnall, et al., 2013). Thirty-one paleomagnetic sites were sampled across the Peak Range province.

2.5. Nebo

Nebo volcano hosts numerous, partly eroded mafic flows, although extensive weathering and lateritization of these units has occurred since Oligocene times preventing paleomagnetic sampling of the rocks in the region. Relatively more resistant silicic shallow intrusives are scattered throughout the area and were more amenable to sampling, albeit being more remote compared with other provinces described here, preventing more extensive sampling. Seven paleomagnetic sites of rhyolitic tuff, flow banded rhyolite, and trachyte plugs were sampled, these being preserved as cliffs and escarpments. Sanidine from a trachyte dyke intruding silicic pyroclastics yielded an $^{40}Ar/^{39}Ar$ age of 32.6 ± 0.4 Ma (Cohen, Knesel, et al., 2013).

2.6. Hillsborough

Hillsborough is the oldest volcano along the Cosgrove hot spot track. An $^{40}Ar/^{39}Ar$ age of 33.6 ± 0.5 Ma from feldspar of at Mount Jukes constrains the age of the volcano (Cohen, Knesel, et al., 2013). The largest remnant of this volcano is exposed at Cape Hillsborough, where a ∼ 350-m-thick series of flows, pyroclastics, and rhyolitic plugs are preserved (McDougall & Slessar, 1972). The remainder of the volcano is more heavily eroded, with its deeper plumbing system, the Mount Jukes Intrusive Complex, now exposed at the surface and with isolated remnants of columnar jointed trachyte also present in the province. Twenty-four paleomagnetic sites were sampled from this province.

3. Rock Magnetic Analysis and Anisotropy of Magnetic Susceptibility

3.1. Susceptibility-Temperature Curves

In order to determine magnetic mineralogy and demagnetization characteristics at each site, small volumes of crushed rock were prepared, weighed, and then loaded into quartz glass holders and magnetic susceptibility changes with temperature (K_m-T curves) measured using an Agico MFK-1 kappabridge with CS-4 furnace. Samples were heated from room temperature to 700 °C and cooled back to room temperature in an inert atmosphere. Samples generally show irreversible behavior the precise nature of which varies between sites of different lithologies and magnetic mineralogies, with examples of both lower and higher final susceptibility than initial values, as well as complex curves related to mineral breakdown and growth of new magnetites (Figure S1). In spite of this, K-T curves are dominated by steeply decaying heating curves with Curie temperatures of ∼ 580 °C, diagnostic of titanium-poor magnetite. Ti-rich magnetites occurred but were more rarely encountered (Figure S1, PR2.2). Curie points > 600 °C were not observed, and hematite is an extremely uncommon magnetic phase. In spite of this, thermal demagnetization of pilot samples from a few sites did reveal small amounts of stable remanence above 580 °C that were demagnetized by 675 °C, suggesting a few cases of hematite (inferred by a reddish coloration of these samples) or maghemite occurring in coexistence with magnetite. A reduction in susceptibility during heating is observed in some samples, suggesting that some alteration is caused by heating. K_m-T curves range from nearly reversible (Figure S1, BV8.4), to examples with dramatic differences, suggesting thermal inversion of more strongly magnetized materials during heating (Figure S1, SV9.4 and PR35.7). Cooling curves with higher final susceptibilities than initial values indicated newly formed magnetite, probably produced by oxy-exolution of titanomagnetite or ilmenite (Figure S1, HV25.5 and HV8.7). Unless pilot thermal demagnetization revealed a remanence above the magnetite Curie point, or alternating field (AF) demagnetization proved ineffective, AF demagnetization was chosen to minimize the detrimental effects that neoformation of magnetite particles might have on isolating characteristic remanence components.

3.2. Anisotropy of Magnetic Susceptibility

Anisotropy of magnetic susceptibility (AMS) ellipsoids were measured on multiple samples from each site, to examine the preservation of flow fabrics and primary flow dips and to evaluate the possibility of any tilting that may have gone unrecognized during fieldwork. Bulk susceptibility (K_m) is positively skewed from 3.65×10^{-5} to 6.29×10^{-2} SI, with significantly higher values in mafic rocks (median = 1.38×10^{-2} SI) than in felsic rocks (median = 0.33×10^{-2} SI, Figure S2). Degrees of anisotropy (P_j) are generally low, and their distribution is positively skewed from isotropic values ($P_j \leq 1.01$) to 1.14 at the most extreme, with a median of 1.02, consistent with the undeformed nature of volcanic material sampled. There is no significant difference in degree of anisotropy between mafic and felsic rocks. The shape of AMS ellipsoids are variable between specimens and range from very prolate ($T = -0.984$; $K_1 - K_3$) to very oblate ($T = 0.986$; $K_1 - K_2$) with

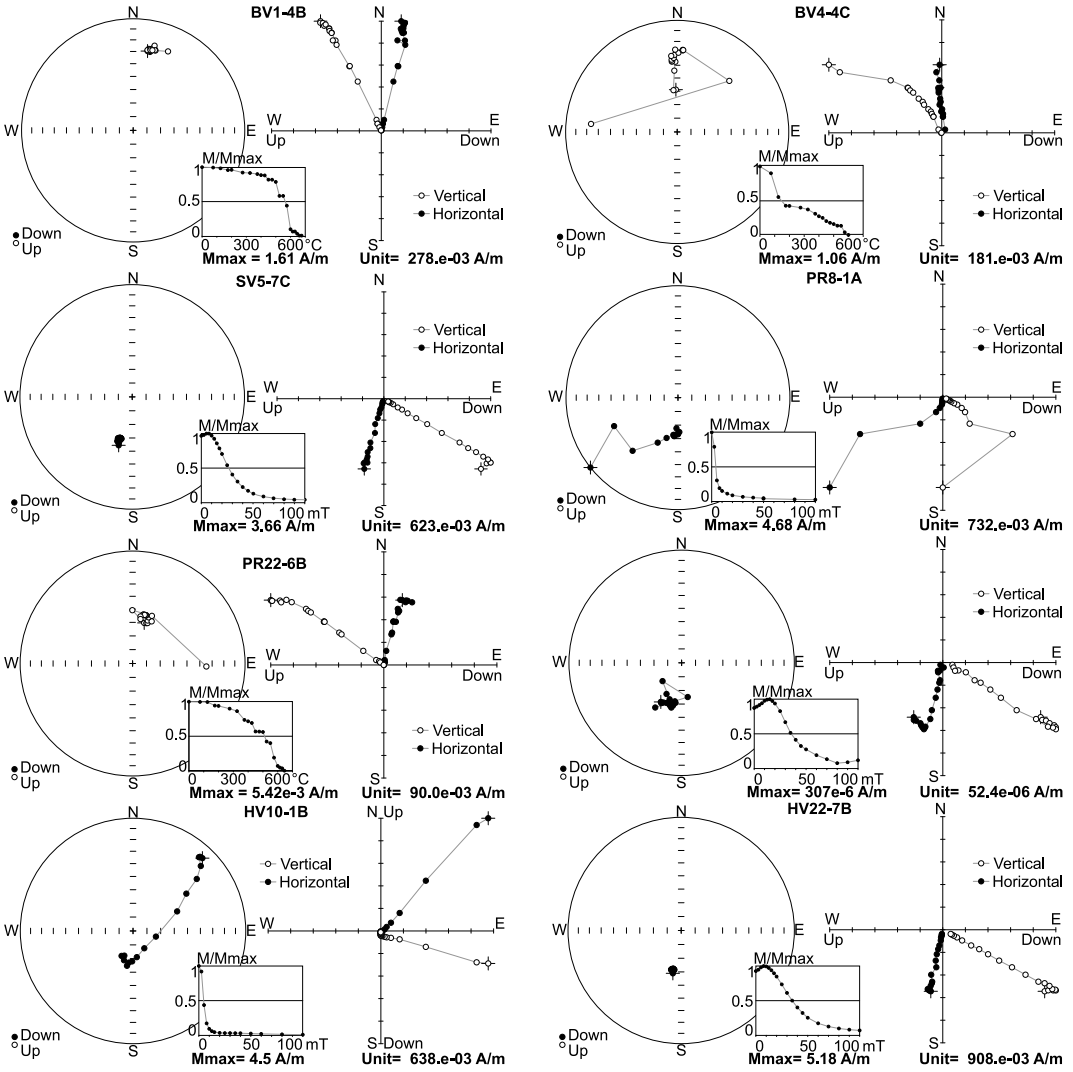


Figure 2. Typical demagnetization behavior for samples from Oligocene Volcanoes along the Cosgrove Hot Spot Track in tectonic/geographic coordinates. BV = Buckland Volcano; SV = Springsure Volcano; PR = Peak Range; NV = Nebo Volcano; HV = Hillsborough Volcano.

values symmetrically distributed about slightly oblate mean values (mean $T_{\text{specimens}} = 0.191$, $\sigma = 0.413$). Site mean averaging suppresses some interspecimen variability (mean $T_{\text{sites}} = 0.174$, $\sigma = 0.267$) and although mafic sites are slightly more oblate than felsic ones (lava flows are also more oblate than intrusive features, which have more variable shape), they are not significantly so (Figure S2).

At the site level principal axes directions often cluster, and differences between volcanic flows and inclined volcanic structures (plugs and dykes) were identified (Figure S2). Volcanic rocks are preserved in their primary orientation in the field, and AMS data reflect this. Lava flows show steeply inclined minimum axes ($K_3 > 60°$) and shallow K_1 and K_2 inclinations and are generally oblate, indicative of preservation of flow along shallow topographic gradients (i.e., primary dips), with possible modification by gravity settling. Sites from intrusive features (dykes and plugs) show shallow and intermediate K_3 inclinations consistent with the contorted columnar jointing and inclined flow banding observed in these rocks. Exceptions in both cases occur; there are examples of inverse AMS fabrics observed for some sites (e.g., site PR11 records a steeply inclined K_1 axis and shallow K_2 and K_3 axes) and in some cases for individual specimens within sites, a phenomenon commonly reported from samples where single-domain magnetite dominates the AMS signal (Tarling & Hrouda, 1993). While preferential AMS fabrics are developed in all but the least aniostropic sites (i.e., $P_j \leq 1.01$), multiple factors likely contribute to the overall AMS fabric at any given site; physical

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Table 1
Site Mean Fisher Statistics From Buckland and Springsure Volcanic Provinces

Site	Dec (°)	Inc (°)	k	a_{95} (°)	n	λ_s (°N)	ϕ_s (°E)	λ_p (°N)	ϕ_p (°E)	A_{95} (°)
Buckland										
BV1	7.8	−30.6	86.96	6.02	7	−24.92	148.06	−78.87	10.5	5.01
BV2	0.9	−39.5	112.69	5.78	6	−24.92	148.06	−87.41	346.38	5.37
BV3	8.8	−40.6	40.93	8.18	8	−24.92	148.06	−81.81	48.11	7.71
BV4	0.1	−39.8	56.98	7.46	7	−24.92	148.06	−87.7	331.06	6.94
BV5	8.6	−41.7	37.15	9.27	7	−24.92	148.06	−82.08	53.28	8.87
BV6	204.5	51.9	52	7.24	8	−24.91	148.03	−67.24	83.32	8.18
BV7	54.7	−73.3	107.65	5.41	7	−24.91	148.04	−39.12	115.36	9.16
BV8	63.2	−75.9	317.77	2.91	8	−24.91	148.05	−34.06	119.15	5.15
BV9	59.8	−75	46.61	8.26	7	−24.91	148.05	−35.92	117.88	14.4
BV10	61.1	−72	79.71	6.29	7	−24.91	148.05	−36.27	111.86	10.42
Springsure										
SV1	142.4	74.7	289.12	3.4	7.5 (15)	−24.24	148.20	−44.95	172.61	5.9
SV2	310.5	28.8	90.87	5.89	7	−24.22	148.20	27.58	92.4	4.81
SV3	29.6	−50.8	138.71	4.76	7	−24.09	148.05	−62.9	80.52	5.27
SV4	355.5	−48.3	34.64	10.51	6	−24.09	148.06	−83.46	184.5	11.15
SV5	185.9	56.1	124.55	6.15	5	−24.09	148.06	−76.51	127.48	7.5
SV6	192.5	60	79.05	6.32	7	−24.09	148.06	−70.23	119.2	8.31
SV7	205.5	59.6	82.21	7.91	5.5 (11)	−24.10	148.04	−63.12	101.5	10.31
SV8	18.7	−60	129.54	4.7	8.5 (1)	−24.08	148.07	−66.64	109.67	6.89
SV9	19.3	−60.1	159.52	4.11	8	−24.08	148.07	−66.65	109.2	5.42
SV10	10.5	−50.3	64.19	8.59	5	−24.08	148.07	−78.45	97.18	9.43
SV11	12.2	−51.2	195.18	4.55	6.5 (3)	−24.08	148.08	−76.71	96.72	5.08
SV13	190.6	42.7	46.37	8.28	7	−24.03	148.11	−80.35	64.05	8.04
SV14	156.6	67.9	182	3.85	8	−24.02	148.12	−57.57	175.98	5.91
SV15	224	72.1	342.18	2.8	8	−24.06	148.15	−44.36	116.31	4.65
SV16	154.2	73.5	1039.72	2.21	5.5 (11)	−24.08	148.10	−50.43	168.49	3.76

Note. Dec = Declination; Inc = Inclination; k = precision parameter (directions); a_{95} = 95% confidence angle (directions); n = number of samples (parentheses indicate number of samples where arc fitting was applied); λ_s, ϕ_s = site latitude and longitude; λ_p, ϕ_p = VGP latitude and longitude; A_{95} = 95% confidence angle (VGPs); VGP = virtual geomagnetic pole.

processes such as convection, gravity settling, and temperature gradient-induced local cooling stresses may modify a primary lava flow fabric, while mineralogical and magnetic domain state influences as well as alteration can also complicate the AMS fabric.

4. Paleomagnetic Analysis

4.1. Demagnetization

Demagnetization and rock magnetic measurements were carried out at the Curtin University Paleomagnetic Laboratory. Magnetic remanence was measured using either an Agico JR6 Spinner Magnetometer or 2G-Instruments vertical cryogenic magnetometer, both of which were housed inside a mu-metal field free room. Progressive AF and thermal demagnetization (at least 17 steps per specimen) was employed using a magnetically shielded furnace (maximum temperature 700 °C), inline AF demagnetizing coils (2G), or a tumbling AF demagnetizer (JR6; maximum field 100 mT) with pilot samples being used to determine best methods for demagnetization at each site. Component magnetizations were determined by principal component analysis (unanchored fits) using at least four demagnetization steps with maximum angular deviation (MAD) < 10° and in cases where simultaneous removal of magnetic components occurred, by the arc method ($n \geq 4$ steps, MAD < 15°) using PaleoMag software (Jones, 2002; Kirschvink, 1980;

Table 2
Site Mean Fisher Statistics From Peak Range Volcanic Province

Site	Dec (°)	Inc (°)	k	a_{95} (°)	n	λ_s (°N)	ϕ_s (°E)	λ_p (°N)	ϕ_p (°E)	A_{95} (°)
Peak Range										
PR1	210.4	50	48.27	8.11	7	−22.40	147.95	−61.62	81.77	8.87
PR2	193.8	34.8	177.46	3.9	8	−22.48	147.91	−76.65	46.1	3.4
PR4	208.9	57.7	101.98	5.6	7	−22.54	147.82	−60.69	97.14	6.99
PR5	343.3	−69.9	78.82	7.19	6.5 (1)	−22.54	147.83	−56.23	165.59	11.45
PR6	5.4	−59.9	102.16	6.08	6	−22.54	147.83	−71.18	135.04	7.97
PR7	1.7	−62.9	60	8.25	6.5 (1)	−22.54	147.83	−68.09	144.57	11.49
PR8	189.9	64.2	52.89	7.18	8	−22.54	147.82	−65.18	131.23	10.25
PR9	263.2	−75.8	143.58	4.68	7	−22.66	147.71	−17.17	175.76	8.26
PR10	233.1	76.6	217.79	3.79	7	−22.70	147.65	−35.9	122.65	6.79
PR11	251.7	50.9	114.34	7.81	4.5 (3)	−22.78	147.79	−26.67	82.76	8.64
PR12	249.1	60.7	53.49	7.7	7	−22.78	147.79	−35.32	94.56	5.75
PR14	196.4	49.3	150.2	4.57	7	−22.67	148.01	−73.57	88.46	4.93
PR15	194.3	6.4	88.4	5.19	9	−22.67	148.01	−66.06	5.55	3.69
PR16	198.9	76.1	223.47	3.75	7	−22.67	148.01	−47.18	135.78	6.64
PR17	178.7	21.8	46.51	9.05	6	−22.67	148.01	−78.57	321.67	6.95
PR18–19	183.9	45.5	55.88	6.52	10 (20)	−22.67	148.01	−84.44	108.84	6.61
PR20–21	75.1	62.4	133.34	3.68	12.5 (3)	−22.67	148.01	−5.48	192.56	5.08
PR22	18.7	−54.3	224.73	4.09	6	−22.67	148.01	−69.62	98.84	4.83
PR23	8.3	−68.1	49.86	9.77	5	−22.67	148.01	−60.79	137.27	15.03
PR24	347.1	−80.5	322.33	3.12	7	−22.67	148.01	−40.45	153.33	5.88
PR25	75.7	−75.1	92.2	5.85	7	−22.82	148.14	−26.66	117.56	10.22
PR26	169.8	31.1	149.01	5.02	6	−22.79	148.17	−78.68	268.58	4.2
PR27	225.9	55.5	137.69	4.74	8 (2)	−22.79	148.22	−48.28	87.39	5.71
PR28	239.4	42.4	229.25	3.43	8	−22.79	148.23	−35.97	72.76	3.31
PR29	221.3	78.5	38.96	8.65	8.5 (1)	−22.81	148.17	−38.32	129.74	15.9
PR30	185.8	78	66.8	6.88	7	−22.81	148.17	−45.62	144.98	12.56
PR31	191	70.2	96.92	5.7	7	−22.81	148.17	−57.51	136.16	9.13
PR32	211.2	48.9	238.43	3.63	7	−22.80	148.17	−61.24	78.99	3.89
PR33	14.1	−54	47.28	8.97	6	−22.48	147.87	−72.75	105.34	10.53
PR35	4.3	−43.5	130.21	5.08	7.5 (1)	−22.48	147.87	−85.1	94.68	4.99
PR36	0.1	−37	35.87	8.2	9	−22.48	147.87	−88.12	331.48	7.35

Note. See Table 1 for information on column labels.

McFadden & McElhinny, 1988). Further tests and calculations were facilitated by the use and expansion of select tools in the PmagPy catalogue (Tauxe et al., 2016). Site mean directions were determined by Fisher statistics (Fisher, 1953), assigning a unit weighting to line fits and a 0.5 weighting to directions fit by the arc method. Example demagnetizations are shown in Figure 2, and all individual fits are described in supporting information Data Set S1. Individual site mean VGPs were calculated where within-site directions were consistent with one another ($k > 30$, $a_{95} < 15°$, $n \geq 5$).

4.2. NRM and Magnetic Components

Natural remanent magnetizations (NRMs) are skewed from expected values for basalts and rhyolites (i.e., 0.1 to ~ 10 A/m, median NRM intensity is 0.624 A/m) toward higher values that are probably the result of isolated lightning overprints (Figure S3). Of 683 specimens, 79 (11%) are greater than 1.5 times the interquartile range ($LQ = 0.17$ A/m, $UQ = 1.6$ A/m) above from the upper quartile. The most extreme magnetization measured was 165.7 A/m. Even without any demagnetization NRM specimen directions are nonrandom and, although scattered, show clear normal and reversed polarities with considerable overlap between

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Table 3
Site Mean Fisher Statistics From Nebo and Hillsborough Volcanic Provinces

Site	Dec (°)	Inc (°)	k	a_{95} (°)	n	λ_s (°N)	ϕ_s (°E)	λ_p (°N)	ϕ_p (°E)	A_{95} (°)
Nebo										
NV1	103.9	74.3	123.16	5.05	7	−21.42	148.60	−25.33	180.36	8.71
NV4	211.3	70.7	39.48	10.22	6.5 (3)	−21.41	148.59	−49.12	121.5	16.51
NV5	209.7	57.1	81.64	6.22	7	−21.41	148.60	−59.68	97.72	7.73
NV6	199.1	53.9	150.08	5.19	6.5 (1)	−21.41	148.60	−68.77	100.44	6.08
NV7	201.9	52	90.32	7.09	6 (2)	−21.41	148.60	−67.54	93.29	8.03
NV12	208.1	66.6	71.01	9.93	4.5 (3)	−21.61	148.61	−54.62	116.41	14.85
NV13	16.6	−25.5	47.63	11.2	5	−21.18	148.52	−74.41	56.04	9.85
Hillsborough										
HV1	6.1	−42.9	37.71	10.46	6.5 (1)	−21.12	148.95	−83.23	94.28	10.18
HV5	355.4	−63.1	171.99	4.85	6.5 (1)	−21.07	149.01	−66.25	157.14	6.77
HV6	176.5	50.1	160.56	4.84	6	−21.00	148.93	−79.61	164.72	5.3
HV7	194	57.1	291.85	3.59	6	−21.00	148.93	−69.32	116.02	4.46
HV8	205	61.9	131.4	4.89	7	−21.01	148.93	−59.61	111.26	6.67
HV9	199.9	62.8	106.53	5.44	7	−21.01	148.93	−61.56	118.14	7.55
HV10	186	66.5	259.37	4.43	5.5 (1)	−21.00	148.93	−61.57	140.65	6.62
HV11	167.4	69.2	127.62	5.95	6 (2)	−21.12	148.86	−56.63	162.73	9.37
HV16–17	210.8	26.9	50.77	5.37	12	−20.92	149.04	−59.92	51.23	4.3
HV18	197.6	63.2	117.35	5.18	7	−20.93	149.06	−62.14	121.64	7.24
HV19	205.7	58	242.99	3.59	7	−20.93	149.06	−61.63	103.64	4.54
HV20	199.4	66.9	924.4	1.84	7	−20.93	149.06	−57.48	124.43	2.77
HV21	207	61.8	229.5	3.7	7	−20.93	149.06	−58.43	109.76	5.03
HV22	194.5	60.8	237	3.98	6	−20.93	149.06	−65.79	122.08	5.31
HV23	93.2	49.7	55.84	7.53	7	−20.93	149.06	−13.12	211.08	8.19
HV24	125.9	0.7	34.6	11.77	5	−20.93	149.06	−33.34	253.19	8.32
HV25	255	73.8	1034.38	1.74	7	−20.92	149.06	−25.46	116.67	2.97
HV26	265.3	70.2	193.29	4.93	5	−20.92	149.06	−19.54	110.91	7.9

Note. See Table 1 for information on column labels.

populations (Figure S3), but NRM directions do not pass a reversal test (McFadden & McElhinny, 1990; Watson, 1983).

Progressive demagnetization generally removed a weak secondary overprint magnetization (s1) at low temperature (< 200 °C) or low peak fields (< 20 mT) revealing a clear characteristic component demagnetizing toward the origin (c1). In five specimens a second overprint component (s2) was identified. At most sites AF demagnetization was effective, and in these cases thermal demagnetization of pilot samples produced directions indistinguishable from AF treatment and was complete before heating to 600 °C, suggestive of magnetite. Thermal demagnetization, or low field AF demagnetization (0–20 mT) followed by thermal demagnetization, was used for sites where a remanence component remained in pilot samples demagnetized with AF alone. In these cases some sites were still effectively demagnetized below 600 °C suggesting magnetite with coercivities higher than the maximum available demagnetizing field (100 mT). In a few sites, a small component of magnetization remained after sharp decreases below 600 °C, indicative of hematite coexisting with magnetite (e.g., BV7-10, PR16, PR24, and NV1) although significant changes in demagnetization vector were not apparent. In sites SV1, SV7, SV16, and PR18–19, as well as for a few isolated specimens within sites, curved demagnetization required the use of arc fits (c1a).

Magnetic overprint components (s1 and s2) are nonrandomly distributed ($R = 264.2, R_0 = 33.9$) and concentrated around the present-day field position (Figure S4), although with considerable scatter, presumably introduced by overlapping characteristic and secondary components, lightning overprints, or other

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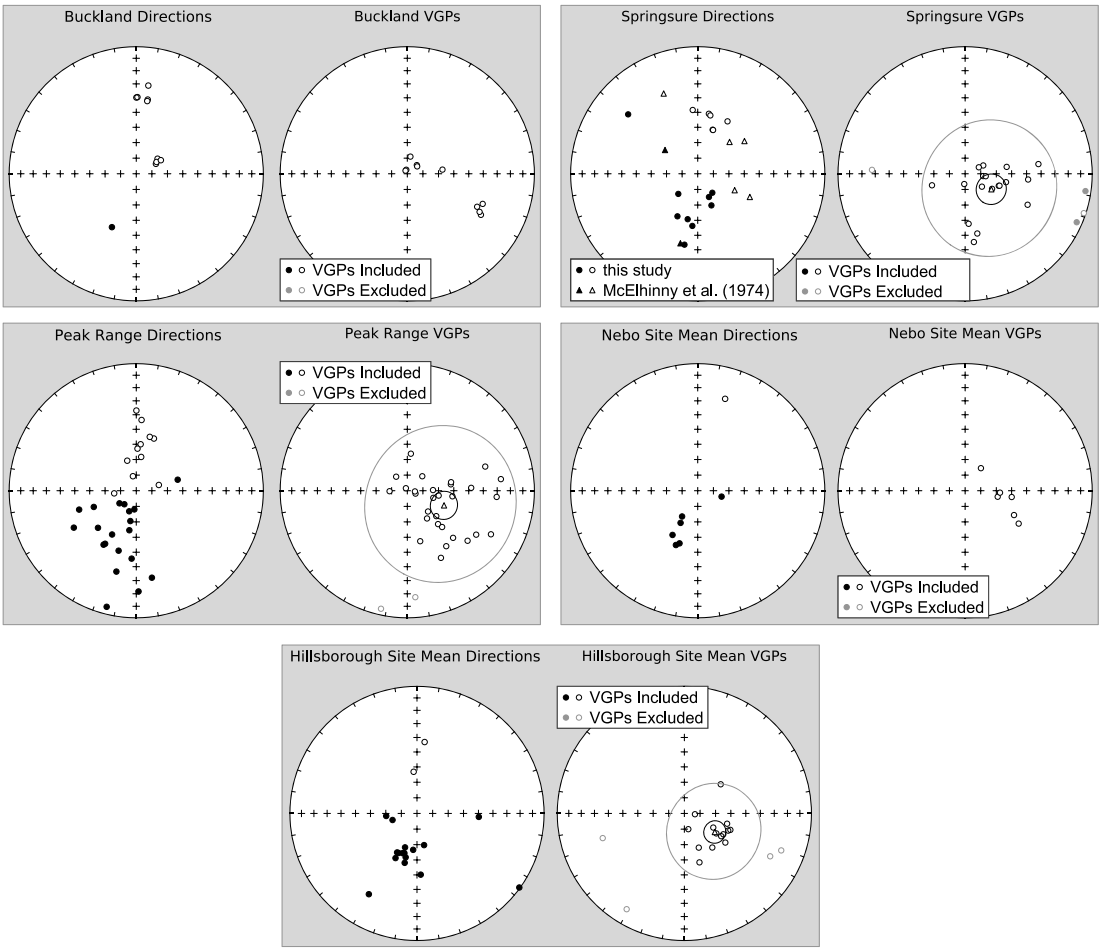


Figure 3. Equal area plots of site mean directions and VGPs for sampled Oligocene volcanoes along the Cosgrove Hot Spot Track in tectonic/geographic coordinates. Gray circles indicate VGP cutoff angle (Vandamme, 1994). All site mean VGPs are described in Tables 1–3. VGP = virtual geomagnetic pole.

secondary effects. The Fisher mean direction of secondary magnetization components is $D = 1.3°$, $I = -65.4°$ ($a_{95} = 5.48$, $k = 2.49$, $n = 441$).

At 81 of the total 94 sites sampled, characteristic magnetizations were sufficiently concentrated ($k > 30$, $a_{95} < 15°$, $n \geq 5$) and site mean results calculated. These directions and their corresponding VGPs are given in Tables 1–3, and plotted for each volcanic province in Figure 3. Of the 11 sites that did not meet this criteria, six (PR34, NV3, HV3, HV4, HV12, and HV13) are randomly distributed according to the criteria of Watson (1956), while the remaining seven sites (PR3, NV2, NV10, NV11, HV2, HV14, and 15) are not randomly distributed but still insufficiently concentrated for an accurate determination of a site mean direction.

4.3. Field Tests, Polarity, and Paleomagnetic Poles From Oligocene Volcanic Provinces of Central Queensland

Sites appropriate for field tests were found at two localities; a conglomerate test was performed at Sprinsure Volcano, and an unconformity test was performed at Hillsborough volcano. At Springsure an intraformational agglomerate of volcanic clasts preserved between lava flows was found and passed ($R = 3, N = 5$). An unconformity test at Cape Hillsborough was conducted by sampling basement, impure carbonate rocks containing shell fragments (Wedge Island Beds), exposed at low tide between Cape Hillsborough and Wedge Island that are overlain by a flow of basalt. The polarity of the basal flow was reversed (Dec/Inc = 255°/73.8°), but the underlying country rock preserve a normal polarity magnetization (Dec/Inc = 13.5°/−53.3°) that is compatible with their middle Eocene age (McNamara, 1994). This, combined with numerous primary Mesozoic and Paleozoic paleomagnetic poles from accreted terranes in eastern Australia into which Cenozoic volcanics are intruded (McElhinny et al., 2003), argues against regional Neogene and

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Table 4
Summary of Published Oligocene Paleomagnetic Data for the Australian Plate

Study/Region	Q	Category	ϕ_p	λ_p	K	A_{95}	S	$S_{95\%l}$	$S_{95\%u}$	N/Total	Age	Type	Ref
Buckland Volcano	5	Igneous								10	26–32	Ar-Ar	1
Springsure Volcano	6	Igneous	300.6	70.5	13.7	9.7	21.8	17.7	28.6	18/22	27.3–28.9	Ar-Ar	2
Peak Range	6	Igneous	291.8	64.6	10.1	8.8	25.4	21.5	31.2	29/31	28.2–31.2	Ar-Ar	3
Nebo	5	Igneous								7	32.2–33	Ar-Ar	4
Hillsborough Volcano	7	Igneous	301.4	67.1	32.2	7.1	14.3	11.2	19.6	14/18	33.1–34.1	Ar-Ar	5
Existing poles													
Main Range Volcano	5	Igneous	271.6	76.6	23.5	6.1	16.7	13.9	20.9	25/29	20.2–26.7	Ar-Ar	6
Tweed Volcano	5	Igneous	306.9	77.4	15.4	5.4	20.6	18.0	24.0	48/50	23.1–24.7	Ar-Ar	7
Pt. Addis Limestone	6	Sedimentary	299.4	68.5	47.8	4.8	11.7	9.6	15.1	20/20	22.4–27.8	Ar-Ar and Biostratigraphic	8
Glasshouse/Noosa	5	Igneous								2	26.2–27.5	Ar-Ar	9
Turritella Clays	6	Sedimentary	291.8	65	110.2	2.4	7.7	6.6	9.4	32/33	32–37	Biostratigraphic	10
Liverpool Volcano	5	Igneous	271	70.9	29.4	4.4	14.9	12.9	17.8	38/48	34–36	K-Ar	11
Combined Oligocene	6	Both	298.2	71.0	24.6	1.9	16.3	15.3	17.4	227/274	29–30	Median/mean age	
Other Cenozoic combined poles													
Plio-Pleistocene	5	Both	353.0	87.7	41.1	3.2	12.6	11.1	14.7	48/51	1–3	Median/mean age	12
mid-Miocene	6	Both	283.6	80.0	26.9	3.0	15.7	14.1	17.4	85/93	15–17	Median/mean age	13
Paleocene	4	Both	301.3	62.6	16.8	4.2	19.7	17.7	22.3	73/74	56–57	Median/mean age	14

Note. Poles given are calculated after a (Vandamme, 1994) filter of all published results. Reliability criteria 1–7 after Van der Voo (1990). Central Province and Older Volcanics (McElhinny et al., 1974; Mumme, 1963) excluded for reasons discussed in text. ϕ_p = pole longitude; λ_p = pole latitude S; $S_{95\%l}$, $S_{95\%u}$ = VGP scatter, with lower and upper 95% confidence bounds. References (paleomagnetic result; age bounds): 1–5 = this study; Cohen, Knesel, et al. (2013). 6 = Robertson (1966), Wellman (1975); Knesel et al. (2008). 7 = Robertson (1966), Wellman (1975); Knesel et al. (2008). 8 = Idnurm (1985); McLaren et al. (2009). 9 = Robertson (1966); Cohen et al. (2007). 10 = Idnurm (1994); Shafik and Idnurm (1994). 11 = Wellman et al. (1969); Vasconcelos et al. (2008). 12 = Barbetti (1977), Barbetti and Sheard (1981), Opdyke and Musgrave (2004); Barbetti (1977), Barbetti and Sheard (1981), Opdyke and Musgrave (2004). 13 = Hansma and Tohver (2018); Hansma and Tohver (2018). 14 = Hill et al. (2002), Idnurm (1985), Robertson (1979), Schmidt and Embleton (1981); Idnurm (1985), Embleton et al. (1985), Vasconcelos et al. (2008).

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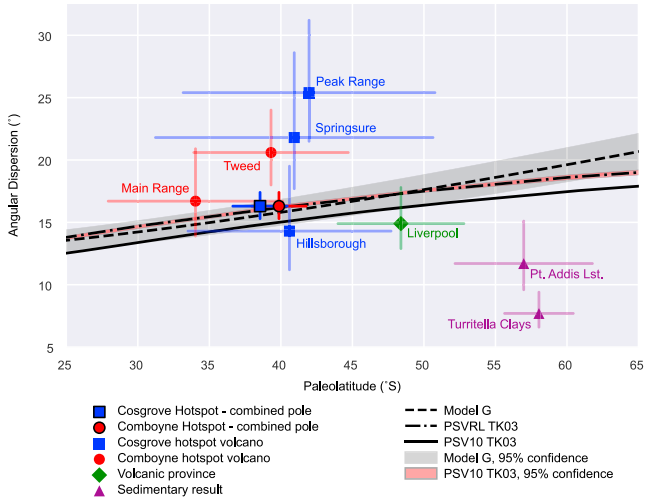


Figure 4. Angular dispersion with paleolatitude for Oligocene paleomagnetic results from Australia, compared with field models derived from databases of recent lavas (0- to 5-Ma lavas of the PSVRL database, Model G, and TK03 shown (McElhinny & McFadden, 1997; Tauxe & Kent, 2004), and TK03 fit of 0- to 10-Ma lavas from the $PSV10_{vcut}$ database (Cromwell et al., 2018).

younger remagnetization events. Dual polarities were found at all five volcanic provinces sampled, andsome sites with transitional VGPs were identified (Vandamme, 1994). Peak Range VGPs pass a reversals test (McFadden & McElhinny, 1990; Watson, 1983) with a C classification ($V = 3.6, V_{crit} = 6.3, angle = 15.1, angle_{crit} = 19.9$). At Springsure a reversal test is also passed, but with an indeterminate classification ($V = 0.7$, $V_{crit} = 6.1, angle = 7.9, angle_{crit} = 23.3$), as do results at Hillsborough ($V = 2.0, V_{crit} = 8.6, angle = 13.9, angle_{crit} = 28.9$).

The ten sites sampled at Buckland volcano form three clear directional clusters (Figure 3). Although the Fisher mean of these VGPs gives a plausible pole position, the sites from this province represent a small number of flows, and paleosecular variation is unlikely to be adequately sampled. Similarly, while the pole position for Nebo sites is plausible, limited outcrop and recovery has resulted in only seven VGPs, too few for a robust paleomagnetic pole. Sufficient success with sites from Springsure ($N = 15$ sites, 22 including data published by McElhinny et al., 1974), Peak Range ($N = 31$ sites), and Hillsborough provinces ($N = 18$ sites) were found, and for these paleomagnetic poles were calculated using a Vandamme (1994) filter to identify transitional VGPs within each province. These poles are presented in Table 4, along with previously published Oligocene poles from Australia, recalculated using the same filtering process used here and with updated geochronologic constraints on magnetization age where appropriate.

Our results from Springsure satisfy all seven Q-score criteria, but common mean testing between VGPs from our sampling and the seven previously published sites (McElhinny et al., 1974) do not reject being drawn from the same distribution. We choose to include these sites and give this pole $Q = 6$. The Peak Range paleomagnetic results satisfy all Q criteria but paleomagnetic field tests. Although Peak Range passes a reversals test with C classification, and positive field tests at Hillsborough and Springsure are also reassuring, we assign this result $Q = 6$ also. Paleomagnetic results at Hillsborough Volcano satisfy all paleomagnetic quality criteria and are given the maximum possible score, $Q = 7$.

4.4. Sampling Paleosecular Variation and Consistency With Geomagnetic Field Models

The 95% confidence limits on VGP scatter for Oligocene results from Hillsborough Volcano compare well with expected values based on 0- to 5-Ma lavas (Paleosecular Variation of Recent Lavas, PSVRL: Model-G and TK03) and 0- to 10-Ma lavas (PSV10:TK03; Cromwell et al., 2018; McElhinny & McFadden, 1997; Tauxe & Kent, 2004), as do existing results from Liverpool Volcano and Main Range (Robertson, 1966; Wellman, 1975; Wellman et al., 1969; Figure 4). This indicates that paleosecular variation has been adequately sampled. Results from Springsure, Peak Range, and Tweed (Robertson, 1966; Wellman, 1975; Wellman et al., 1969) volcanoes show slightly higher than expected VGP scatter. This could be due to either the influence of transitional directions in the results or a meaningful difference between Oligocene geomagnetic field scatter and < 10 Ma field scatter on which models are based. Central Province results (McElhinny et al., 1974) are highly scattered ($S = 50.5°$), possibly due to blanket demagnetization incompletely removing overprints, a preponderance of transitional results, or other unidentified problematic effects, and for this reason we exclude them from further calculations. Unlike volcanic Oligocene results, those from sediments show lower than expected VGP scatter, possibly as a consequence of some PSV averaging within the sedimentary beds (Idnurm, 1985, 1994).

A Fisherian distribution of VGPs implies that site-level paleomagnetic directions will be distributed elliptically with long axis oriented N-S. The elongation of this ellipse varies by latitude, decreasing from low latitudes

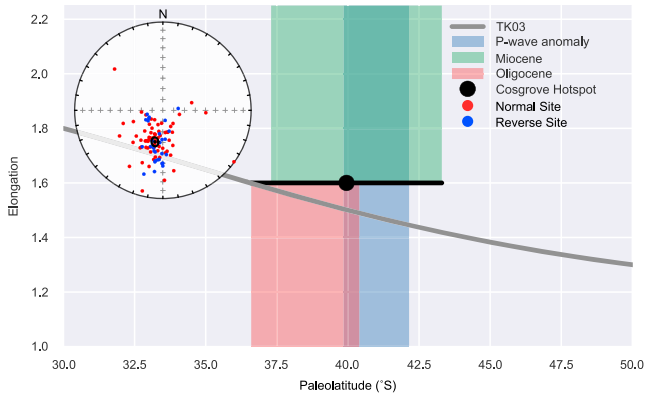


Figure 5. Elongation versus paleolatitude for Cosgrove hot spot lavas compared to predicted values based upon TK03 (Tauxe & Kent, 2004) with equal area stereonet showing site mean directions (gray) and their Bingham mean (black). Shading indicates paleolatitude bounds for the hot spot during the Oligocene (red: this study), Miocene (green: Hansma & Tohver, 2018), and today (blue: *P* wave anomaly limits, Montelli et al., 2006).

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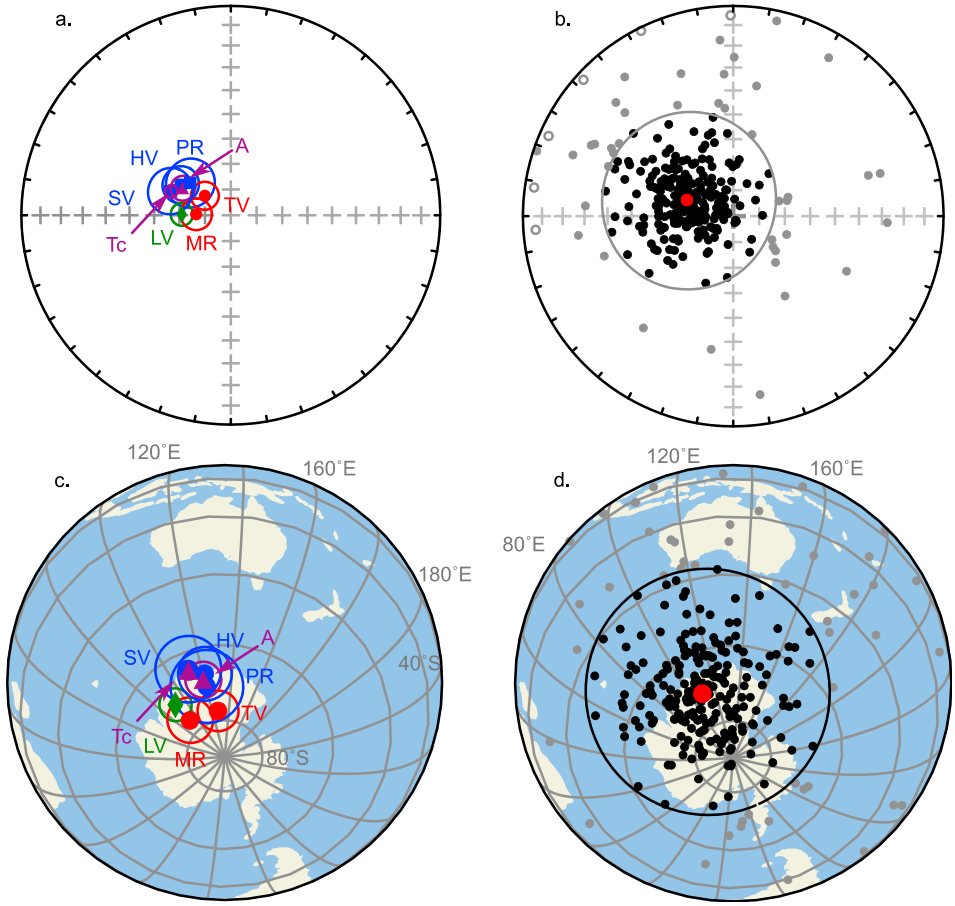


Figure 6. (a) Equal area plot of Australian Oligocene poles (north); red circles indicate Comboyne hot spot volcanics, blue squares indicate Cosgrove hot spot volcanics, green diamond indicates lava fields, and purple triangles indicate sedimentary results. (b) Equal area plot of combined Oligocene pole with included (black) site mean VGPs and excluded (gray) VGPs outside Vandamme cutoff. (c) Orthographic plot of (south) pole data from individual studies; coloring follows panel (a) above. (d) Orthographic plots of combined Oligocene pole and site mean VGPs. SV = Springsure Volcano; PR = Peak Range; HV = Hillsborough Volcano; MR = Main Range; TV = Tweed Volcano; LV = Liverpool Volcano; Tc = Turritella Clays; A = Point Addis Limestone; VGP = virtual geomagnetic pole.

to high (Tauxe & Kent, 2004). As Oligocene paleolatitudes for the Cosgrove hot spot are indistinguishable from those during mid-Miocene times (Hansma & Tohver, 2018), volcanic rocks from this hot spot represent recordings of the magnetic field from a single location on Earth's surface between 34 and 9.8 Ma. Including younger, mid-Miocene results from leucitite lavas of the Cosgrove hot spot, 97 sites record magnetic directions from the same region on Earth's surface. A Bingham mean of magnetic vectors from these sites provides empirical validation of the TK03 geomagnetic field model, having a long axis of elongation with an azimuth 354° and elongation of $E = 1.6$. This is remarkably similar to the expected N-S azimuth and elongation of 1.4–1.6 predicted by TK03 geomagnetic field model at Cosgrove paleolatitudes (Figure 5).

4.5. Common Mean Testing Between Studies and Combined Oligocene Poles for Australia

We justify combining Oligocene VGP data into a composite pole because of the coincidence of most poles with one another and the partial overlaps in magnetization age across these studies. For each individual study there are some small differences in pole position (Figure 6); for instance, Liverpool and Main Range are not significantly different from one another (despite being of Late and Early Oligocene, respectively), yet they are distinct from Turritella, Hillsborough, Pt. Addis Lst., Peak Range, and Springsure results. Furthermore, the Fisher mean directions all fall within one circular standard deviation of each other, except for the most concentrated result from the Turritella clays, where within site PSV averaging is artificially enhancing the concentration of VGP data. Although this effect confounds the meaningful comparison of Turritella VGPs with others, means from other studies fall within two circular standard deviations of even this result.

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There is no obvious pattern between mean position and age for each result, so these small differences in mean position probably have arisen because of small biases in sampling secular variation within each study. The combined pole position of 289.2°E, 71.0°N, $K = 24.6$, $A_{95} = 1.9°$, $N = 227/274$, $Q = 6$, (excluding Central Province and Older Volcanics VGPs for aforementioned reasons) has VGP scatter of 16.3° (15.3°, 17.4°: 95% confidence limits) and falls within expected limits of PSV models. We consider this combined pole to better reflect an Oligocene pole position for Australia than any individual Oligocene pole might, due to its more comprehensive sampling of PSV. The combined pole includes VGP data with magnetization ages from 20 to 36 Ma that all overlap with the Oligocene interval, and the mean and median VGP age are 29.3 and 29.6 Ma, respectively. As such a reconstruction age of 29–30 Ma is most appropriate for this pole.

The combined Oligocene pole has VGPs consistent with a Fisher distribution ($Me = 0.98$, $Me_{\text{crit}} = 1.094$, $Mu = 0.83$, $Mu_{\text{crit}} = 1.207$, $N = 227$). This is a somewhat unexpected result, as these VGPs represent PSV, combined with ~ 11 Ma of northward Australian plate motion. VGPs collected over PSV timescales, that is, 10^4 to 10^5 years (or at extremes ~ 5 Ma) are often assumed to be Fisher distributed (Tauxe & Kent, 2004; Tauxe et al., 2018), and including apparent polar wander would introduce elongation. It does not appear that there are strong biases toward sampling around a narrow interval during the Oligocene. Synthetic APWPs for Australia, based upon seafloor spreading isochrons, can be generated from global plate models and indicate between 7.0° (Seton et al., 2012) and 7.5° (Matthews et al., 2016) of APW over the Oligocene interval. It may simply be that over this interval's duration, and given the magnitude of paleosecuar variation at middle latitudes, too little APW occurs to reject a Fisher distribution even at this (albeit high) level of sampling.

5. Australia's APWP During the Cenozoic

We wished to see if continental paleomagnetic data were compatible with evidence from seafloor spreading for the Australian Plate. Of particular interest to us was the acceleration of the Australian plate northward, recorded after anomaly C13 (~ 34 Ma) in the Southern Ocean (Tikku & Cande, 1999). Existing APWPs for Australia fall short in this regard due to bias introduced by erroneous filtering (McElhinny et al., 1974), by over-rejection of available data (Idnurm, 1985, 1994), or by applying a monotonic rate of APW (Musgrave, 1989).

To generate a Cenozoic APWP for Australia, we recalculated several more combined poles for Australia during the Cenozoic, using the same site mean VGP level approach described above (Table 4). A Plio-Pleistocene pole was calculated using existing results from (Barbetti, 1977; Barbetti & Sheard, 1981; Opdyke & Musgrave, 2004). The mid-Miocene combined pole presented in Hansma and Tohver (2018) was calculated with the same approach so we use this pole. Eocene times younger than Ypresian (48–34 Ma) are poorly represented by paleomagnetic data from Australia, with only seven published VGPs (one from Nerriga, four from the Woy Woy Basalt, and two from Peat's Ridge), and we choose not to calculate a pole with such sparse data (Dickson, 1963; Embleton et al., 1985; Manwaring, 1963; McElhinny et al., 1974; Robertson, 1979). Our oldest paleomagnetic pole is of Late Paleocene-Early Eocene age (50–60 Ma) and combines results from Barrington Volcano by Hill et al. (2002), whose results corroborate but supercede earlier blanket demagnetization by Wellman et al. (1969), with results from North Rankin 1 (Idnurm, 1985) and Mogo Hill Basalt (Robertson, 1979; Schmidt & Embleton, 1981). In all cases we take the median VGP age, as opposed to the less robust but more commonly applied mean age, to be the magnetization age of the pole.

Our new APWP is very different from an early proposal based on igneous-only results (McElhinny et al., 1974); notably the new path does not feature large swings of this earlier APWP (Figure 7A). The APWP presented by Idnurm (1985, 1994; Figure 7b), rejected most igneous results in favor of sedimentary ones but included blanket demagnetized Newer Volcanics Province results by Aziz-Ur-Rahman (1971) in the 2.9-Ma pole. Opdyke and Musgrave (2004) resampled Newer Volcanics Province results with stepwise demagnetization and obtained a significantly different VGP sample to the earlier result. The rejection of igneous data is baseless, particularly given the coincidence of igneous and sedimentary results as demonstrated by common mean testing of VGP distributions between igneous and sedimentary studies of similar age presented in section 4.5 and in Hansma and Tohver (2018). Furthermore, VGP scatter indicates adequate sampling of PSV in these igneous studies (section 4.4), contrary to suggestions by Idnurm (1985) that they are inadequate in this regard. As a consequence of this mode of data selection, bias was introduced into the sedimentary APWP (Idnurm, 1985, 1994) that overestimates the rate of Australian plate motion during much of the Cenozoic, especially between 26 Ma and today (Figure 7B). The opposite case occurs for the APWP of Musgrave (1989;

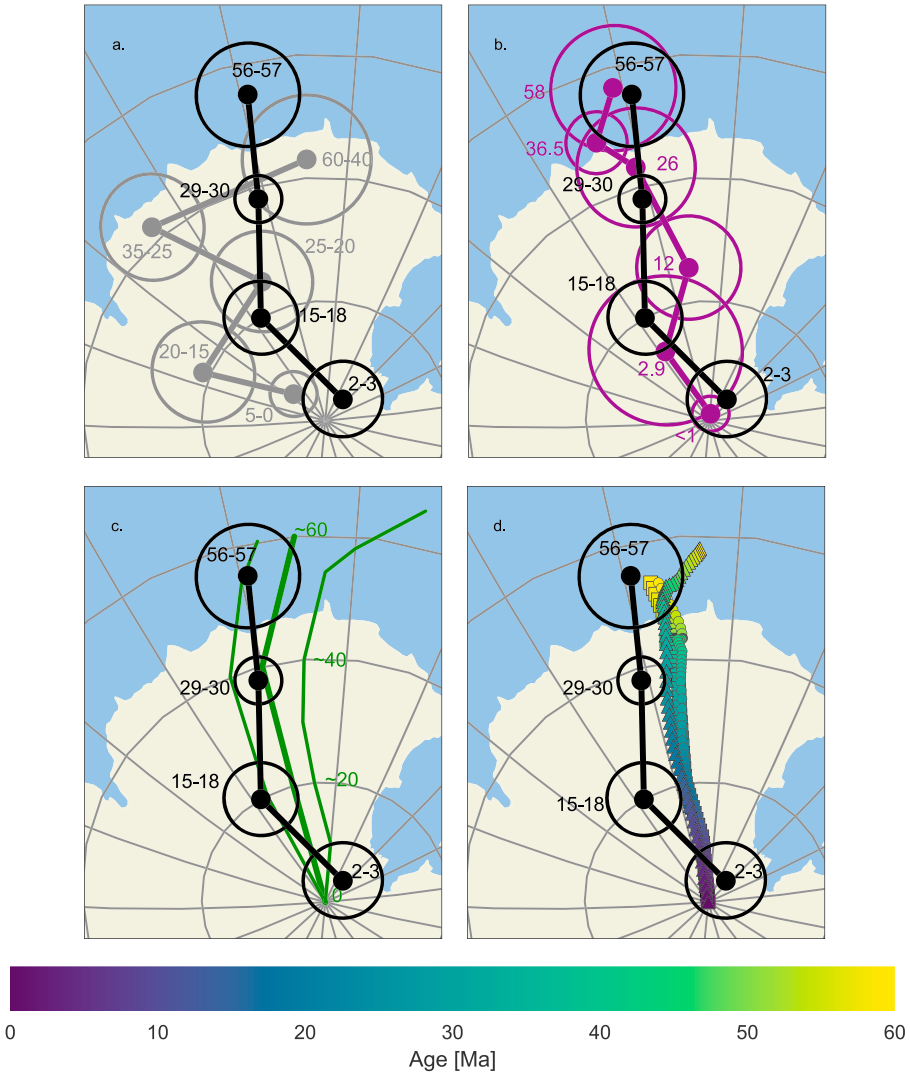


Figure 7. Australia's combined pole Cenozoic APWP (black) compared with (a) early igneous poles (gray; McElhinny et al., 1974), (b) sedimentary results (purple; Idnurm, 1985, 1994), (c) igneous and laterite data with a monotonic spline (green; Musgrave, 1989), and (d) synthetic APWPs based on seafloor-spreading models of Seton et al. (2012; squares), Matthews et al., (2016; triangles), and Whittaker et al. (2013; circles). APWP = apparent polar wander paths.

Figure 7c), which implies a constant rate of seafloor spreading. In this case, the smoothed age progression assigned to poles masks abrupt changes in drift rate that are evident from the underlying data.

The newly proposed APWP reproduces the key elements of a synthetic path based on global reconstructions of seafloor magnetic anomalies, namely, correspondence of shape and the predicted age for pole locations, although models predict pole longitudes slightly east of observed paleomagnetic pole positions (Figure 7d). It is likely, however, that uncertainties in seafloor magnetic fits would overlap with continental pole data. Pole latitudes are remarkably consistent with those predicted from seafloor models (Matthews et al., 2016; Seton et al., 2012; Whittaker et al., 2013) and an increase in APW (in the continental APWP) from 0.3°/Ma between ~ 56–29 Ma to 0.6° afterward is consistent with seafloor-spreading models that document Australia's northward acceleration after anomaly C13 (Seton et al., 2012). Anomaly C13 marks the clearance of the South Tasman Rise from the East Antarctic continental margin, changing this boundary from continental strike-slip to oceanic transform (Cande & Stock, 2004). Acceleration may have occurred earlier, in the middle Eocene (Whittaker et al., 2007, 2013), but this is not resolved by available paleomagnetic data. Notably, however, plate rotations associated with a NE-SW early separation between Australia and Antarctica inferred by Whittaker et al. (2007) and incorporated in Matthews et al. (2016) are not supported by current

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continental Paleocene paleomagnetic data from Australia. While Whittaker et al. (2007) point out that the rotations (Tikku & Cande, 1999) used in the Seton et al. (2012) model produce unacceptable overlaps between the South Tasman Rise and East Antarctica, unrealistic episodes of extension and compression of the Kerguelen Plateau present in both models led to further revision (Whittaker et al., 2013). The Whittaker et al. (2013) model resolves both aforementioned fit problems and is also compatible with Australian continental paleomagnetic data.

Detecting changes in rates of plate motion using paleomagnetic data is ultimately dictated by VGP scatter and by geological preservation. It is well known that VGP scatter caused by PSV is greater than continental motion over short timescales, but over the long-term tectonic drift also matters. As it is assumed that VGPs are independent of one another and as there are simply a limited number of magmatic units and sedimentary beds on a continent for any given period of time, geological preservation ultimately limits paleomagnetic precision, setting a practical limit on what might feasibly be observable using paleomagnetic techniques. Spatial and temporal biases introduced into APWPs also can contribute to obscuring those changes that may be observable in the VGP data that underlie them. Positional bias introduced by averaging poles rather than their underlying VGP distributions can bias position, while common APWP averaging methods can also obscure tectonic changes present in the underlying data. Greater spline smoothing factors increase age decoupling between the APWP and the underlying VGP distribution (Van der Voo, 1993), while in moving-window approaches, by assuming a pole age to be the center of an interval, rather than the more robust median VGP age within that interval, a similar decoupling can also be introduced.

6. Conclusions

We have presented paleomagnetic results from five Oligocene volcanic provinces of the Cosgrove hot spot track in central Queensland and compared these with the body of already published Oligocene paleomagnetic data. The good agreement between results obtained with modern techniques and older results, both in direction and VGP scatter, validate many existing results, with only results from Central Province (McElhinny et al., 1974) and Older Volcanics (Mumme, 1963) being suspect in this regard. We show that magnetic vectors from 97 sites along the Cosgrove hot spot are consistent with expected distributions based on the TK03 geomagnetic field model (Tauxe & Kent, 2004). We presented a combined Oligocene pole based on site-mean VGP level data and combined it with pole data from the rest of the Cenozoic era, confirming that continental paleomagnetic data from Australia reflect a change in rate of motion, compatible with accelerated seafloor spreading in the Southern Ocean after anomaly C13 (~ 34Ma).

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